Low-Level Design (LLD) — CredVault

System: CredVault Credit Card Management Platform
Version: 1.0
Date: April 2026

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1. Design Principles

1.1 Clean Architecture

Every microservice follows a strict four-layer structure:

```
┌──────────────────────────────────────────┐
│              API Layer                   │  Controllers, Middleware, DI setup
├──────────────────────────────────────────┤
│          Application Layer               │  Commands, Queries, Handlers, Validators
├──────────────────────────────────────────┤
│        Infrastructure Layer              │  EF Core DbContext, Repositories, Messagi
├──────────────────────────────────────────┤
│            Domain Layer                  │  Entities, Enums, Domain Events
└──────────────────────────────────────────┘
```

Dependency Rule: Inner layers never depend on outer layers. The Domain layer has zero external dependencies.

1.2 CQRS with MediatR

- **Commands** mutate state (create, update, delete)
- **Queries** read data (fetch, list, search)
- Each handler is isolated — no shared logic between reads and writes
- FluentValidation runs as a pipeline behavior before handlers execute

1.3 Saga Pattern for Distributed Transactions

Cross-service operations (bill payment involves Payment → Billing → Card services) use a **choreography-based saga** via MassTransit State Machine:

- Each step publishes an event; the next step reacts to it
- On failure, compensation events reverse completed steps
- State is persisted so sagas survive service restarts

1.4 Shared Contracts Library

`Shared.Contracts` is a .NET class library referenced by all services providing:

- `BaseApiController` — standardized response methods
- `ApiResponse<T>` — consistent API envelope
- `ExceptionHandlingMiddleware` — global error handling
- `ServiceCollectionExtensions` — JWT, Swagger, MassTransit setup
- All event interfaces and enum definitions

2. Service Architecture

2.1 Identity Service (:5001)

Domain Entities: `IdentityUser`

Commands:

Command	Handler Action
`RegisterUserCommand`	Create user (PendingVerification), generate OTP, publish `IUserRegistered` + `IUserOtpGenerated`
`LoginUserCommand`	Validate credentials, check status, return JWT
`GoogleLoginCommand`	Validate Google IdToken, create/login user, return JWT
`VerifyEmailOtpCommand`	Validate OTP, mark user Active, return JWT
`ResendVerificationCommand`	Regenerate OTP, publish `IUserOtpGenerated`
`ForgotPasswordCommand`	Generate reset OTP, publish `IUserOtpGenerated`
`ResetPasswordCommand`	Validate OTP, update password hash
`UpdateUserProfileCommand`	Update FullName
`ChangePasswordCommand`	Validate old password, set new hash
`AdminUpdateUserStatusCommand`	Change user status (Active/Suspended)
`AdminUpdateUserRoleCommand`	Change user role (User/Admin)
Queries:	
Query	Returns
`GetUserProfileQuery`	Current user profile
`AdminListUsersQuery`	Paginated user list
`AdminGetUserQuery`	Single user details
`AdminGetUserStatsQuery`	Aggregated user statistics
Events Published: `IUserRegistered`, `IUserOtpGenerated`	

2.2 Card Service (:5002)

Domain Entities: `CreditCard` , `CardTransaction` , `CardIssuer`
Commands:

Command	Handler Action
`AddCardCommand`	Encrypt card number, generate masked number, create card, publish `ICardAdded`
`UpdateCardCommand`	Update cardholder name, credit limit, billing cycle
`DeleteCardCommand`	Set `IsDeleted = true` (soft delete)
`SetDefaultCardCommand`	Mark card as default, unset previous default
`AddCardTransactionCommand`	Record purchase/payment/refund
`AdminUpdateCardCommand`	Admin card update
`AddIssuerCommand`	Create card issuer
`UpdateIssuerCommand`	Update issuer details
`DeleteIssuerCommand`	Delete issuer
Queries:	
Query	Returns
---	---
`GetCardsByUserQuery`	User's cards (excludes soft-deleted)
`GetCardByIdQuery`	Single card details
`GetCardTransactionsQuery`	Paginated card transactions
`GetAllUserTransactionsQuery`	All transactions across user's cards
`ListIssuersQuery`	All card issuers
`AdminListCardsQuery`	All cards (including soft-deleted)
Events Published: `ICardAdded`	

2.3 Billing Service (:5003)

Domain Entities: `Bill` , `Statement` , `StatementTransaction` , `RewardAccount` , `RewardTier` , `RewardTransaction`

Commands:

Command	Handler Action
`GenerateBillCommand`	Aggregate transactions, calculate totals/min-due, create bill + statement
`CheckOverdueBillsCommand`	Scan bills past due date, mark as Overdue
`PayBillCommand`	Update bill status, record payment amount
`CreateStatementCommand`	Generate statement from card transactions
`EarnRewardsCommand`	Calculate points from tier rate, add to reward account
`RedeemRewardsCommand`	Deduct points, create RewardTransaction (Redeemed)
`ReverseRewardsCommand`	Restore points, create RewardTransaction (Reversed) — saga compensation
`CreateRewardTierCommand`	Create reward tier
`UpdateRewardTierCommand`	Update tier configuration
`DeleteRewardTierCommand`	Delete reward tier
Queries:	
Query	Returns
---	---
`GetBillsByUserQuery`	User's bills (paginated)
`GetBillByIdQuery`	Single bill details
`HasPendingBillQuery`	Boolean + bill details if exists
`GetStatementsByUserQuery`	User's statements
`GetStatementByIdQuery`	Statement with full breakdown
`GetRewardAccountQuery`	Reward balance + tier info

Command	Handler Action
`GetRewardTransactionsQuery`	Reward history
`ListRewardTiersQuery`	All reward tiers

2.4 Payment Service (:5004)

Domain Entities: `Payment` , `Transaction` , `UserWallet` , `WalletTransaction` , `PaymentOrchestrationSagaState` , `RazorpayWalletTopUp`

Commands:

Command	Handler Action
`InitiatePaymentCommand`	Validate bill, check wallet balance, create Payment, generate OTP, publish `IPaymentOtpGenerated`
`VerifyPaymentOtpCommand`	Validate OTP, publish `IStartPaymentOrchestration` (triggers saga)
`ResendPaymentOtpCommand`	Regenerate OTP
`TopUpWalletCommand`	Create Razorpay order
`DebitWalletCommand`	Deduct wallet balance — saga step
`RefundWalletCommand`	Restore wallet balance — saga compensation
`ProcessRazorpayWebhookCommand`	Verify signature, update wallet, mark top-up complete
Queries:	
Query	Returns
`GetPaymentByIdQuery`	Payment details
`GetPaymentTransactionsQuery`	Payment transaction history

Command	Handler Action
`GetWalletQuery`	Full wallet info
`GetWalletBalanceQuery`	Balance only
`GetWalletTransactionsQuery`	Wallet transaction history
Events Published: `IPaymentOtpGenerated` , `IStartPaymentOrchestration` , saga state events	
Background Jobs: Payment expiration cleanup	

2.5 Notification Service (:5005)

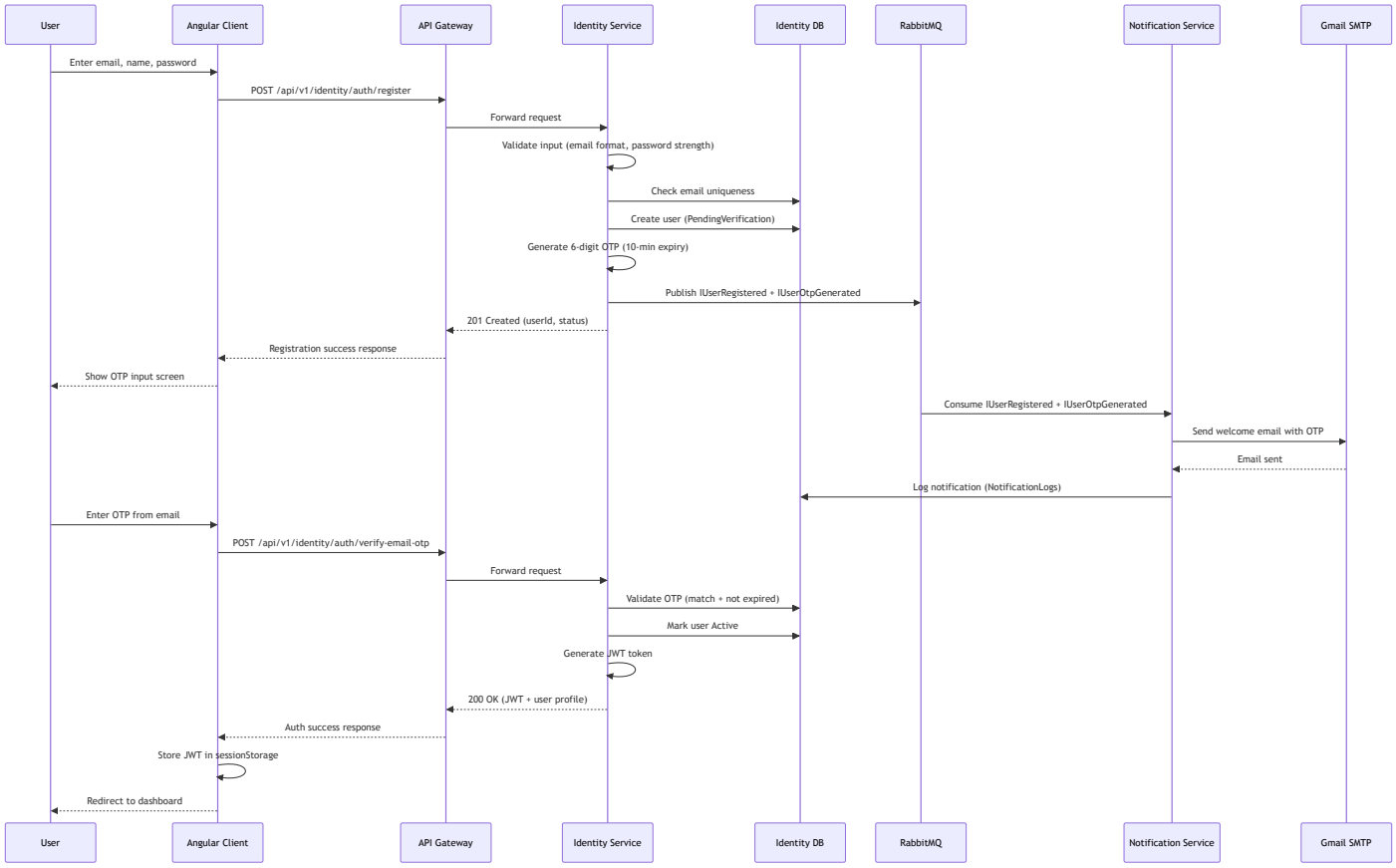
Domain Entities: `AuditLog` , `NotificationLog`
Consumers (MassTransit):

Event Consumer	Action
`IUserRegistered`	Send welcome email
`IUserOtpGenerated`	Send OTP email (verification or password reset)
`ICardAdded`	Send card confirmation email
`IPaymentOtpGenerated`	Send payment OTP email
`IPaymentCompleted`	Send payment success email
`IPaymentCompensated`	Send payment failure email
Commands:	
Command	Handler Action
`SendEmailCommand`	Send via Gmail SMTP, log result
`LogAuditCommand`	Create audit log entry
`LogNotificationCommand`	Create notification log entry

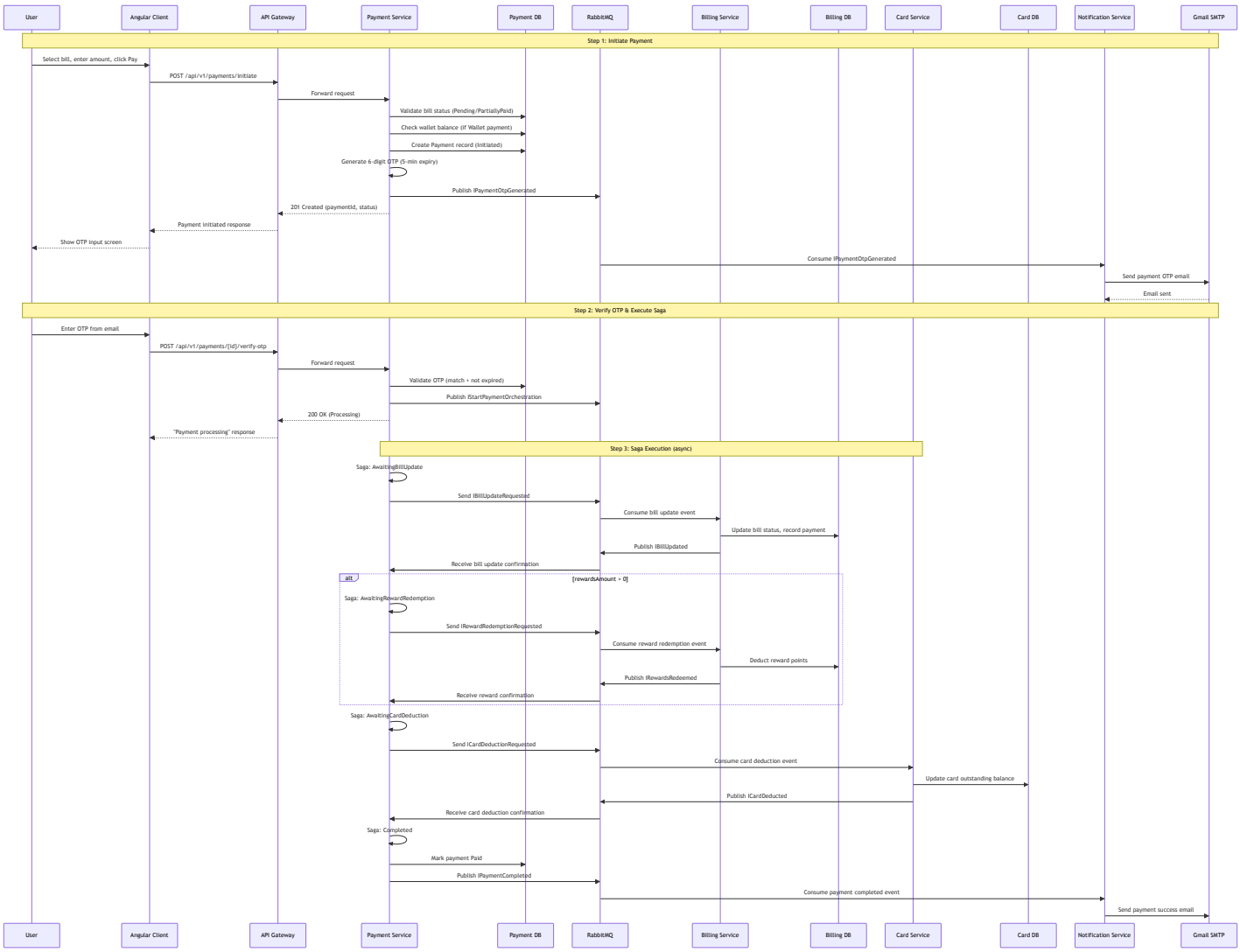
Event Consumer	Action
Queries:	
Query	Returns
---	---
`GetNotificationLogsQuery`	Paginated notification logs (admin)
`GetAuditLogsQuery`	Paginated audit logs (admin)

3. Sequence Diagrams

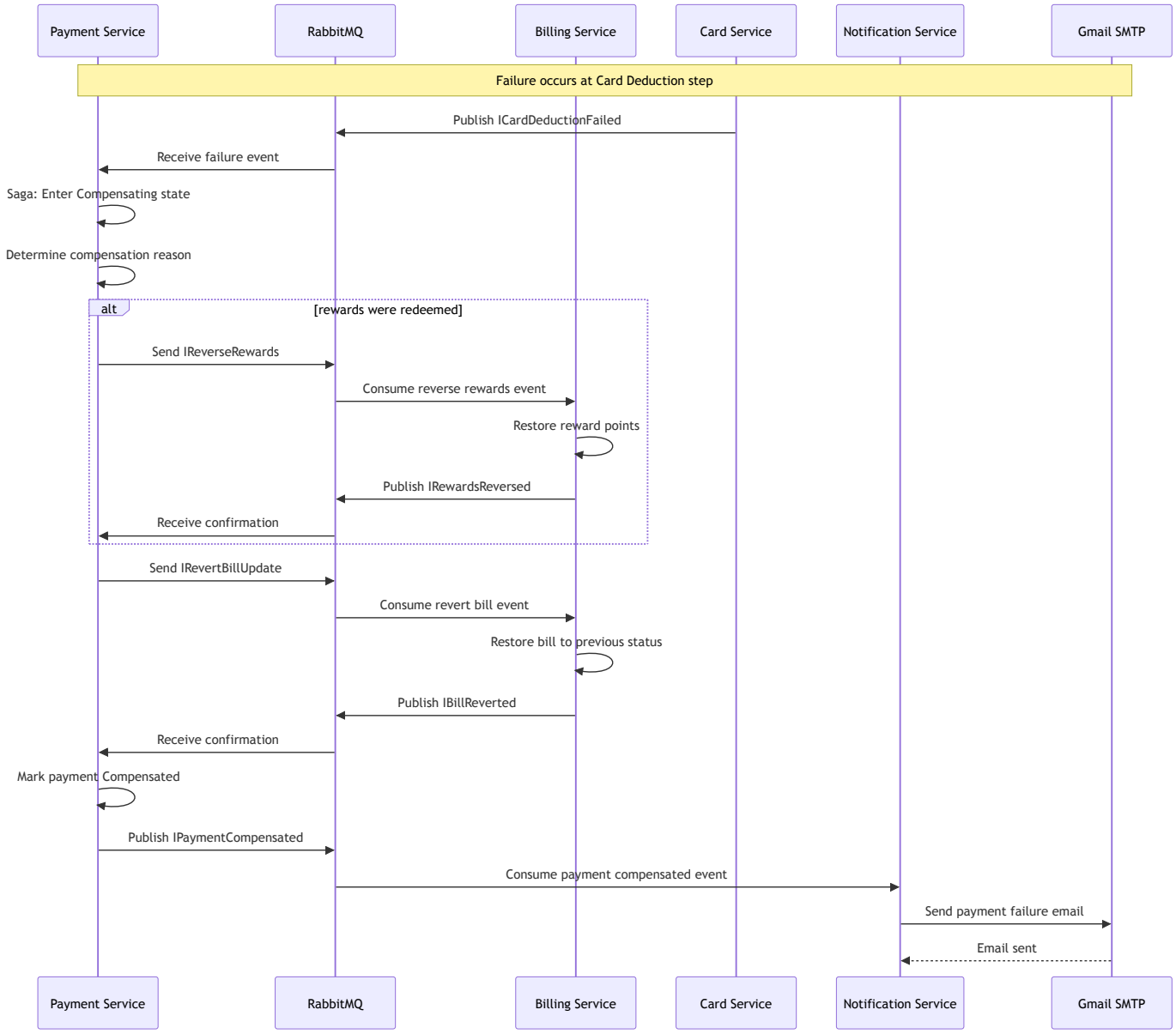
3.1 User Registration & Email Verification



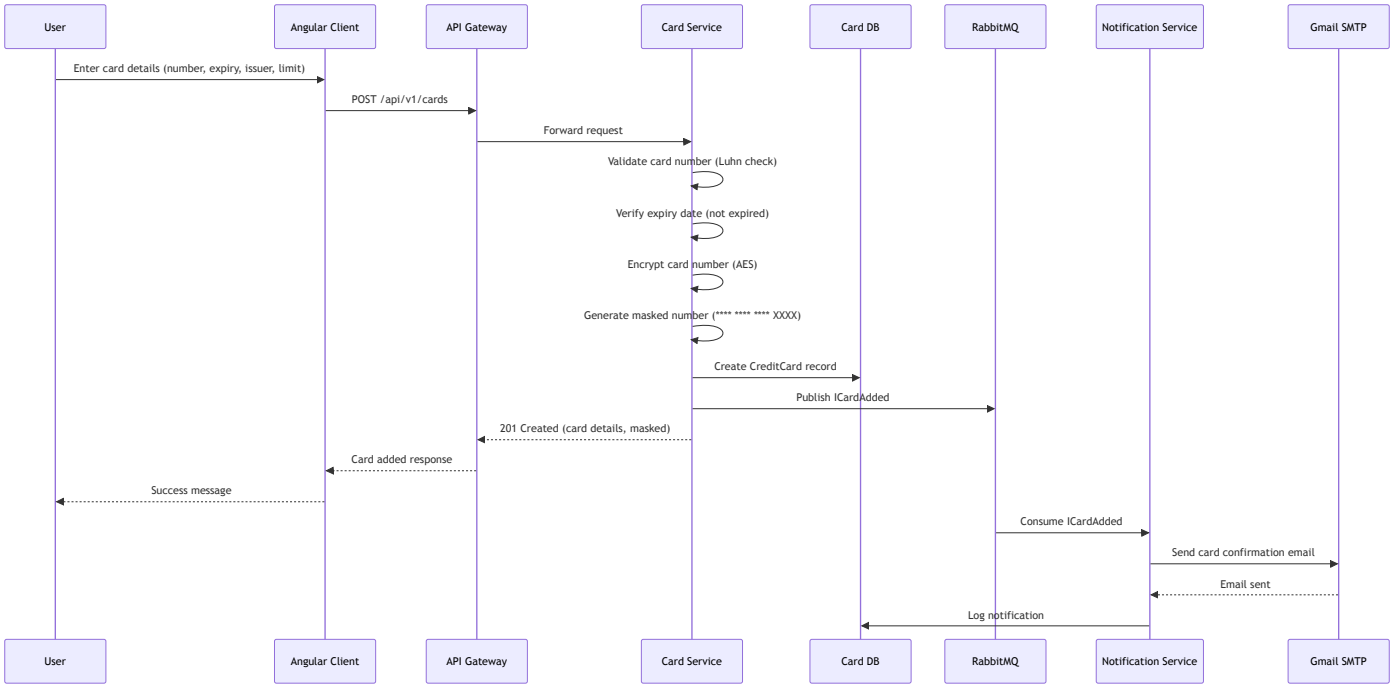
3.2 Bill Payment with Saga Orchestration



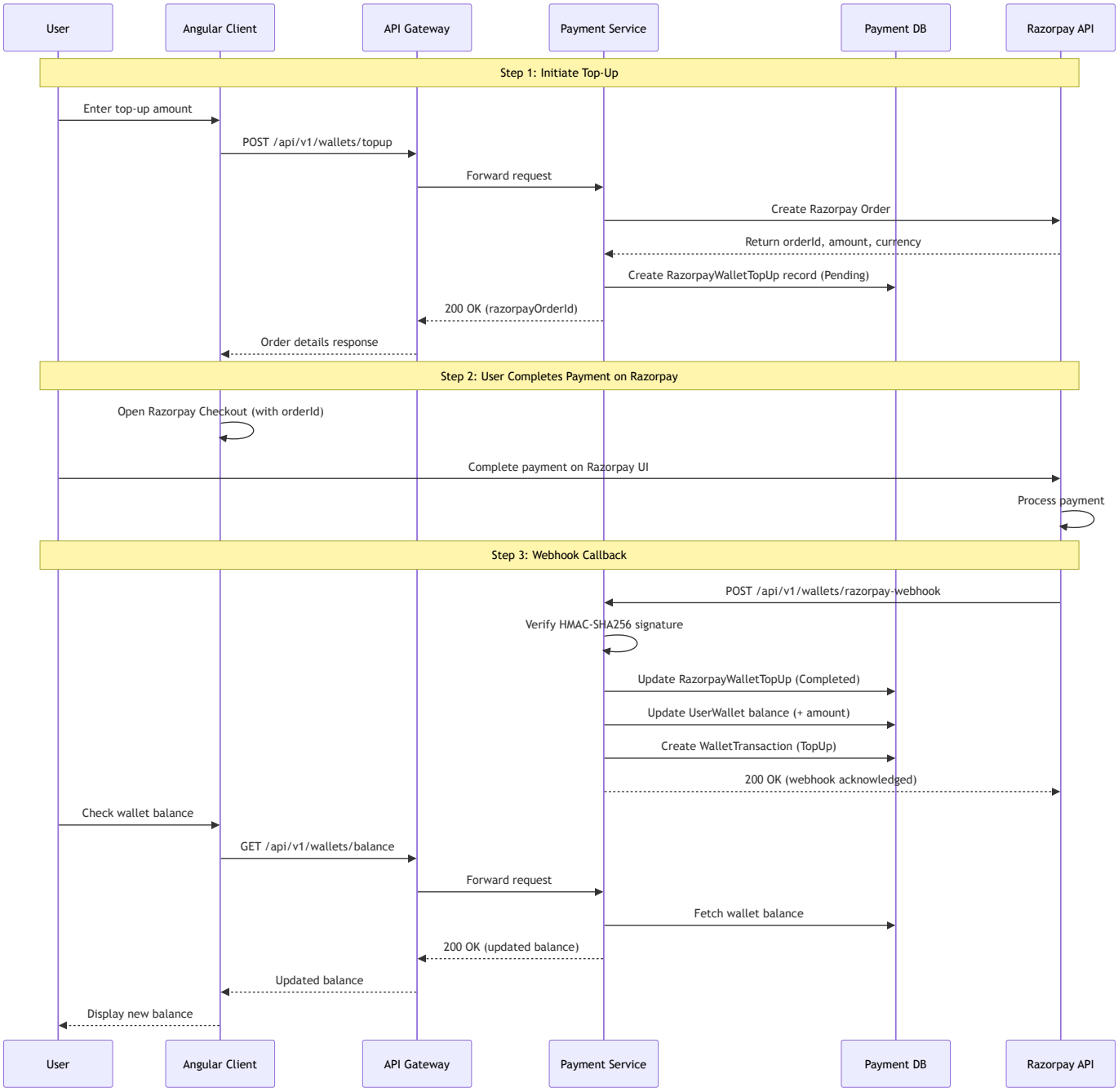
3.3 Saga Compensation (Payment Failure)



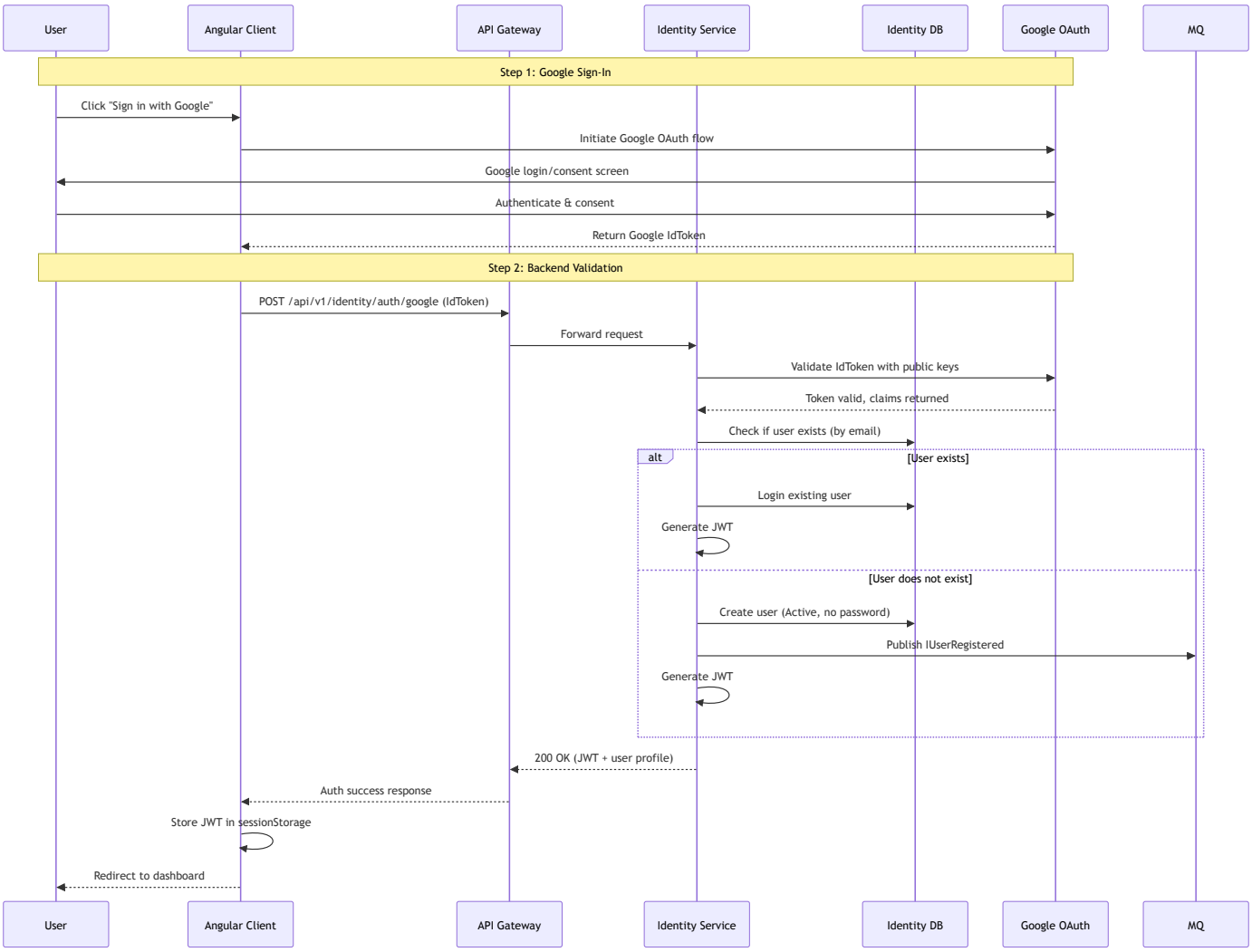
3.4 Card Addition Flow



3.5 Wallet Top-Up via Razorpay



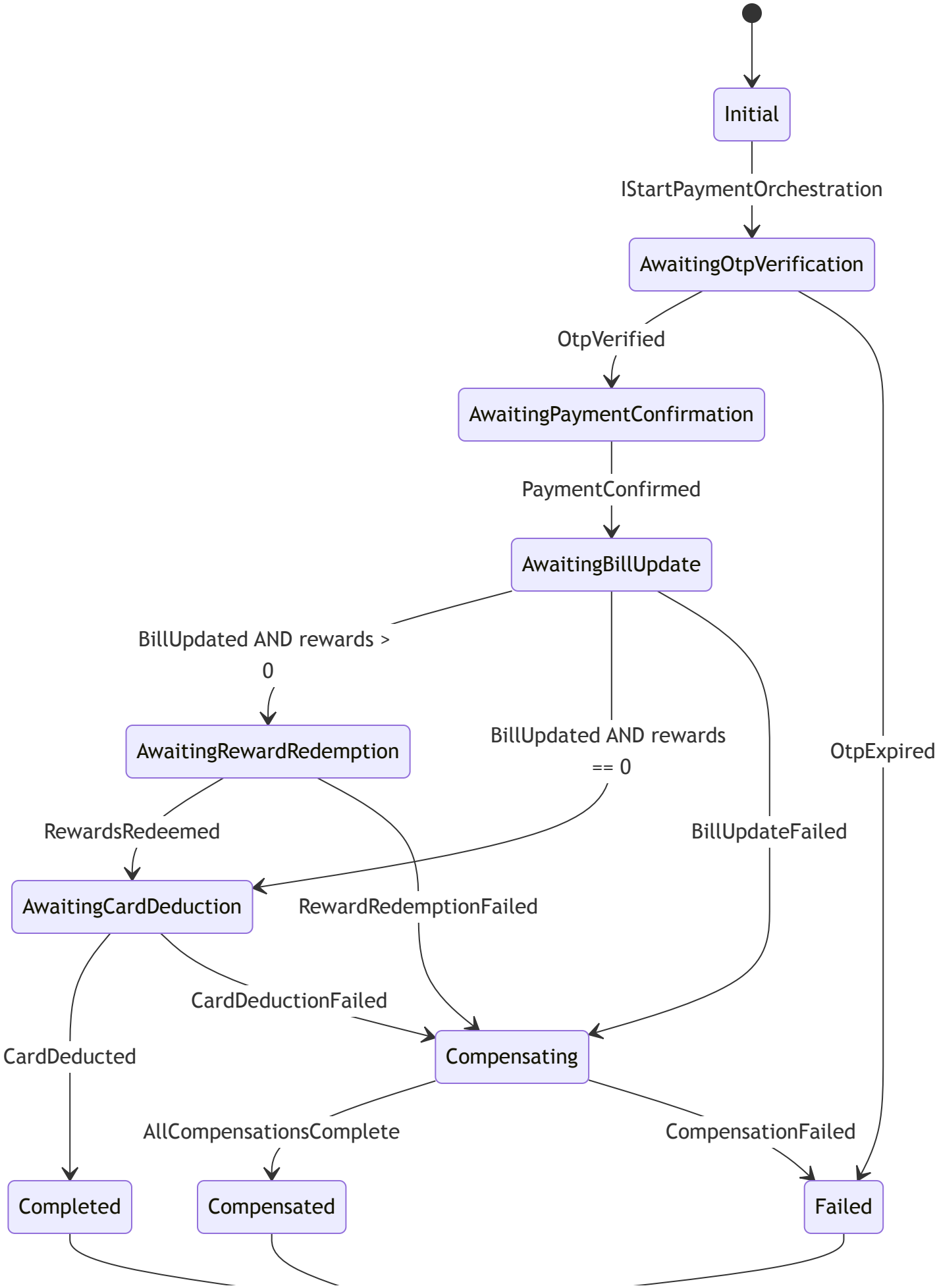
3.6 Google OAuth Login Flow



4. Saga Orchestration

4.1 Payment Saga State Machine

The Payment Service uses a MassTransit State Machine to orchestrate distributed bill payments across Billing and Card services.


Initial
IStartPaymentOrchestration
AwaitingOtpVerification
OtpVerified
AwaitingPaymentConfirmation
PaymentConfirmed
AwaitingBillUpdate
BillUpdated AND rewards > 0
BillUpdated AND rewards == 0
OtpExpired
AwaitingRewardRedemption
RewardsRedeemed
BillUpdateFailed
AwaitingCardDeduction
RewardRedemptionFailed
CardDeductionFailed
CardDeducted
Compensating
AllCompensationsComplete
CompensationFailed
Completed
Compensated
Failed

4.2 Saga Steps & Compensation

Step	Action	Target Service	Compensation
1	Update bill (mark Paid/PartiallyPaid)	Billing Service	Revert bill status
2	Redeem reward points (if applicable)	Billing Service	Reverse points to reward account
3	Deduct card outstanding balance	Card Service	Restore card balance
4	Debit wallet (if wallet payment)	Payment Service	Refund wallet balance

4.3 Saga Reliability

Mechanism	Configuration
Outbox	`UseInMemoryOutbox()` — prevents lost messages during restart
Retry	Exponential backoff: 1s → 5s → 15s (3 attempts)
Timeout	30 seconds per step; triggers compensation
Idempotency	CorrelationId (GUID) as saga PK; duplicates ignored for completed sagas
Persistence	`PaymentOrchestrationSagas` table tracks current state

5. Database Schema

5.1 credvault_identity

Table	Key Columns	Relationships
`identity_users`	Id (PK), Email (UQ), FullName, PasswordHash, IsEmailVerified, EmailVerificationOtp, PasswordResetOtp, Status, Role, CreatedAtUtc, UpdatedAtUtc	—

5.2 credvault_cards

Table	Key Columns	Relationships
`CardIssuers`	Id (PK), Name, Network (enum), CreatedAtUtc, UpdatedAtUtc	1 → N CreditCards
`CreditCards`	Id (PK), UserId (indexed), IssuerId (FK), CardholderName, Last4, MaskedNumber, EncryptedCardNumber, ExpMonth, ExpYear, CreditLimit, OutstandingBalance, BillingCycleStartDay, IsDefault, IsVerified, IsDeleted, CreatedAtUtc, UpdatedAtUtc	N → 1 CardIssuers; 1 → N CardTransactions
`CardTransactions`	Id (PK), CardId (FK), UserId, Type (enum), Amount, Description, DateUtc	N → 1 CreditCards

5.3 credvault_billing

Table	Key Columns	Relationships
`Bills`	Id (PK), UserId, CardId, CardNetwork, IssuerId, Amount, MinDue, Currency, BillingDateUtc, DueDateUtc, AmountPaid, PaidAtUtc, Status (enum), CreatedAtUtc, UpdatedAtUtc	1 → N Statements; 1 → N RewardTransactions
`Statements`	Id (PK), UserId, CardId, BillId (FK), StatementPeriod, PeriodStartUtc, PeriodEndUtc, OpeningBalance, TotalPurchases, TotalPayments, TotalRefunds, PenaltyCharges, InterestCharges, ClosingBalance, MinimumDue, AmountPaid, Status, CardLast4, CardNetwork, IssuerName, CreditLimit, AvailableCredit, CreatedAtUtc, UpdatedAtUtc	N → 1 Bills; 1 → N StatementTransactions

Table	Key Columns	Relationships
`StatementTransactions`	Id (PK), StatementId (FK), SourceTransactionId, Type, Amount, Description, DateUtc	N → 1 Statements
`RewardAccounts`	Id (PK), UserId (UQ), RewardTierId (FK), PointsBalance, CreatedAtUtc, UpdatedAtUtc	N → 1 RewardTiers; 1 → N RewardTransactions
`RewardTiers`	Id (PK), CardNetwork, IssuerId (nullable), MinSpend, RewardRate, EffectiveFromUtc, EffectiveToUtc, CreatedAtUtc, UpdatedAtUtc	1 → N RewardAccounts
`RewardTransactions`	Id (PK), RewardAccountId (FK), BillId (FK, nullable), Points, Type (enum), CreatedAtUtc	N → 1 RewardAccounts; N → 1 Bills

5.4 credvault_payments

Table	Key Columns	Relationships
`Payments`	Id (PK), UserId, CardId, BillId, Amount, PaymentType (enum), Status (enum), FailureReason, OtpCode, OtpExpiresAtUtc, CreatedAtUtc, UpdatedAtUtc	1 → N Transactions
`Transactions`	Id (PK), PaymentId (FK), UserId, Amount, Type (enum), Description, CreatedAtUtc, UpdatedAtUtc	N → 1 Payments
`UserWallets`	Id (PK), UserId (UQ), Balance, TotalTopUps, TotalSpent, CreatedAtUtc, UpdatedAtUtc	1 → N WalletTransactions
`WalletTransactions`	Id (PK), WalletId (FK), Type (enum), Amount, BalanceAfter, Description, RelatedPaymentId, CreatedAtUtc	N → 1 UserWallets

Table	Key Columns	Relationships
`PaymentOrchestrationSagas`	CorrelationId (PK), CurrentState, UserId, CardId, BillId, Email, FullName, Amount, RewardsAmount, PaymentType, CompensationReason, error fields	—
`RazorpayWalletTopUps`	Id (PK), UserId, Amount, RazorpayOrderId (UQ), RazorpayPaymentId (UQ), RazorpaySignature, Status, FailureReason, CreatedAtUtc, UpdatedAtUtc	—

5.5 credvault_notifications

Table	Key Columns	Relationships
`AuditLogs`	Id (PK), EntityName, EntityId, Action, UserId, Changes (JSON), TraceId, CreatedAtUtc	—
`NotificationLogs`	Id (PK), UserId, Recipient, Subject, Body, Type (enum), IsSuccess, ErrorMessage, TraceId, CreatedAtUtc	—

5.6 Cross-Database Constraint Policy

No cross-service foreign key constraints. Each database is fully isolated. Referential integrity across services is maintained through:

- Eventual consistency via RabbitMQ events
- Saga compensation for distributed transactions
- UserId/CardId/BillId passed as plain GUIDs in events and commands

6. Event Contracts

6.1 Identity Events

```
interface IUserRegistered {
  userId: string;
  email: string;
  fullName: string;
  role: string;
  registeredAt: Date;
}
interface IUserOtpGenerated {
  userId: string;
  email: string;
  otp: string;
  otpType: string;  // "EmailVerification" | "PasswordReset"
  expiresAt: Date;
}
```

6.2 Card Events

```
interface ICardAdded {
  cardId: string;
  userId: string;
  cardLast4: string;
  cardNetwork: string;
  issuerName: string;
  addedAt: Date;
}
```

6.3 Payment Events

```typescript
interface IPaymentOtpGenerated {
  paymentId: string;
  userId: string;
  email: string;
  otp: string;
  amount: number;
  expiresAt: Date;
}
interface IStartPaymentOrchestration {
  correlationId: string;
  paymentId: string;
  userId: string;
  cardId: string;
  billId: string;
  email: string;
  fullName: string;
  amount: number;
  rewardsAmount: number;
  paymentType: string;  // "Wallet" | "Card"
}
```

6.4 Saga Events

```
interface IBillUpdateRequested {
  correlationId: string;
  billId: string;
  amount: number;
  paymentId: string;
}
interface IBillUpdated {
  correlationId: string;
  billId: string;
  success: boolean;
}
interface IRewardRedemptionRequested {
  correlationId: string;
  userId: string;
  rewardsAmount: number;
  billId: string;
}
interface IRewardsRedeemed {
  correlationId: string;
  pointsRedeemed: number;
  success: boolean;
}
interface ICardDeductionRequested {
  correlationId: string;
  cardId: string;
  amount: number;
}
interface ICardDeducted {
  correlationId: string;
  cardId: string;
  success: boolean;
}
interface IPaymentCompleted {
  correlationId: string;
  paymentId: string;
  completedAt: Date;
}
interface IPaymentCompensated {
  correlationId: string;
  paymentId: string;
  compensationReason: string;
```

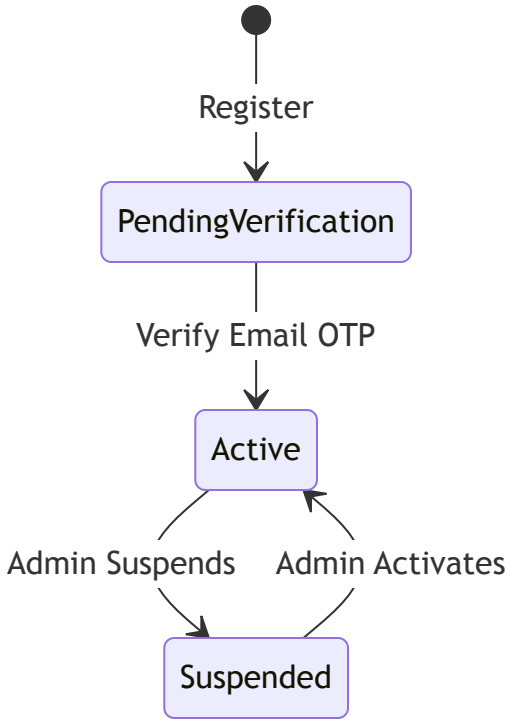
```
  compensatedAt: Date;
}
```

6.5 Event Routing

Event	Publisher	Consumer(s)
`IUserRegistered`	Identity	Notification
`IUserOtpGenerated`	Identity	Notification
`ICardAdded`	Card	Notification
`IPaymentOtpGenerated`	Payment	Notification
`IStartPaymentOrchestration`	Payment	Payment (Saga)
`IBillUpdateRequested`	Payment (Saga)	Billing
`IBillUpdated`	Billing	Payment (Saga)
`IRewardRedemptionRequested`	Payment (Saga)	Billing
`IRewardsRedeemed`	Billing	Payment (Saga)
`ICardDeductionRequested`	Payment (Saga)	Card
`ICardDeducted`	Card	Payment (Saga)
`IPaymentCompleted`	Payment (Saga)	Notification
`IPaymentCompensated`	Payment (Saga)	Notification

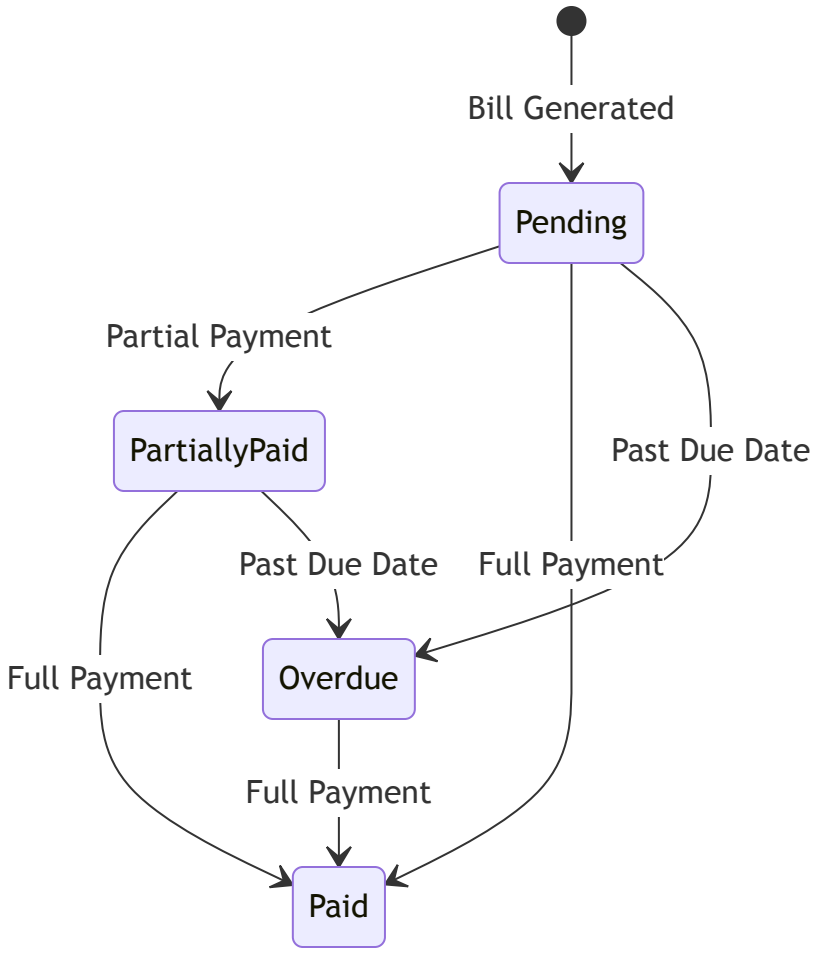
7. State Machines

7.1 User Status



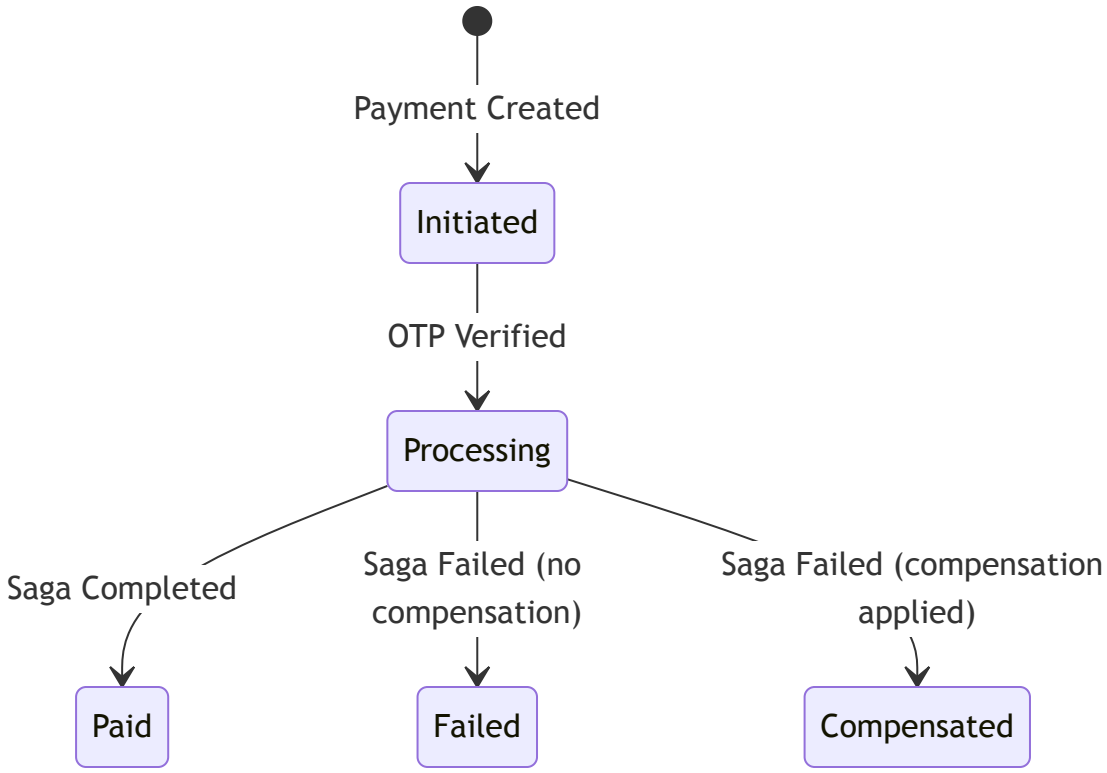
Status	Description
`PendingVerification`	Just registered, awaiting email OTP verification
`Active`	Verified and can use all features
`Suspended`	Disabled by admin; cannot login

7.2 Bill Status



Status	Description
`Pending`	Bill generated, no payments received
`PartiallyPaid`	Some amount paid, balance remaining
`Paid`	Fully settled
`Overdue`	Past due date without full payment

7.3 Payment Status



Status	Description
`Initiated`	Payment record created, OTP sent
`Processing`	OTP verified, saga executing
`Paid`	All saga steps completed successfully
`Failed`	Saga failed, no steps to compensate
`Compensated`	Saga failed, all completed steps reversed

8. Business Rules

8.1 User Rules

- Email must be unique across all users
- Password minimum: 8 characters, 1 uppercase, 1 lowercase, 1 digit
- OTP expires after 10 minutes (5 minutes for payment OTP)
- Suspended users cannot login or perform any action

- Google SSO users have null PasswordHash

8.2 Card Rules

- Card number encrypted before storage (AES)
- Only last 4 digits stored in plain text
- Soft delete (IsDeleted flag) — no physical deletion
- Deleted cards excluded from all queries
- One card can be marked as default; setting a new default unsets the previous

8.3 Billing Rules

- Minimum due calculated as percentage of total bill amount
- Bills transition to Overdue if not paid by due date
- Partial payments allowed (status → PartiallyPaid)
- Statements aggregate all transactions within billing period

8.4 Rewards Rules

- Reward rate determined by tier (network + issuer specific)
- Rewards earned on payment completion
- Rewards redeemable during payment (reduces payment amount)
- Reward redemption is reversible via saga compensation
- Each user has exactly one reward account

8.5 Payment Rules

- Payments require OTP verification (2FA)
- Payment via Wallet or Card
- Wallet balance must be sufficient for wallet payments
- Payment OTP expires after 5 minutes
- Expired payments cleaned up by background job
- All payments go through saga orchestration

8.6 Wallet Rules

- Each user has exactly one wallet (auto-created on first use)
- Wallet balance cannot go negative
- Wallet transactions are immutable

- Razorpay top-ups verified via HMAC-SHA256 signature
- Duplicate webhooks handled idempotently

8.7 Saga Rules

- Saga is idempotent (same CorrelationId = same saga instance)
- Compensation triggered on any step failure
- Compensation executes in reverse order
- Saga state persisted for recovery
- 30-second timeout per step triggers compensation

End of Low-Level Design Document